

Chuck loved me some quantum physics and apparently so does everybody else especially when we're talking about quantum entanglement I'll say some of the best questions we received on quantum entanglement. Yeah, one of the world's experts We went down under to Canberra Australia Sean Hodgman coming right up a physicist on top of the situation

Welcome to Star Talk your place in the universe where science and pop culture collide Star Talk begins right now. This is Star Talk Cosmic Queries Edition What is the subject? Quantum entanglement Chuck are you ready for this? I never

I'm just gonna be honest haven't you and I been a quantum entangled for a while. Yes without a doubt Anytime you feel pain Neil. I feel it immediately That's how that works, you see

Now I have like a storybook understanding of quantum entanglement so to really get to the bottom of this We combed the world to find somebody who actually works in the field And we found a physicist at the Australian National University A and U Research School of Physics there and that would be Sean Hodgman Sean welcome to Star Talk. Thanks a lot for having me. Yeah, this is this is everybody

nobody doesn't like quantum entanglement everybody's into it and We have a million questions before we even get to cause the queries part of this episode We have questions of our own. I have questions of my own and so let's just Let's just come right out of the box and tell me What you publish papers on what what is it you do?

Yeah, so I my my group works on a whole range of experiments Our particular apparatus that we work on involves making helium atoms are really cold So we take them and we cool them down to almost absolute zero So absolute zero is as cold as you can get when there's essentially no motion in the system Again remembering that thermal temperature is basically just random thermal motion

We take all the motion out of the system and we make it really cold the temperatures. We get to our A millionth of a degree above absolute zero. I'd

say that's cold

So listen man, you're you're almost there, okay I Got it keep trying keep at it man. Keep at it. You're almost there man Helium it was familiar as a gas that you can inhale out of a balloon in a in a birthday party Remind me that liquefies around three degrees. Is that correct? So normally at normal pressure it would but because we do it in a vacuum system

We keep it in the gaseous phase. So it's still it's still a gas that these really cold temperatures Just a really low density Whoa, cool Okay, and so why the hell do you do?

We ask ourselves that question too sometimes

Yeah, so the the purpose of this at these cold temperatures all the atoms will form a single coherent quantum state Called a Bose-Einstein condensate. So that's where

quantum mechanically Low temperatures so really cold and when atoms are moving really slowly they don't They don't behave like these little billiard ball situations that we like to think of What they actually behave like is they become these fuzzy smear-out quantum blobs and at these temperatures They all become essentially an identical quantum state, which is very similar to a laser

Where a laser is the same for photons Is just the same thing as as you cool it down its effective wavelength increases Exactly. Yeah, it's not a fair way to say that and so that the wavelength is so long They're all just sharing the same wavelength and so they all they have a hive mind at that point Is that a fair way to characterize this? Definitely So the reason we don't see these quantum effects of particles behaving like waves at normal temperatures is because their wavelength is too small

To actually see once you get to the temperatures that we cool them down to then they form the single quantum state and their wavelengths are actually It's it's macroscopic. So it's inside the trap. It's maybe a hundred micro meters So point one of a millimeter. Um, I'm not great on inches. So if you're American

But yeah, and when we when we drop the atoms from the trap and release them onto our detector So they fall nearly a meter and in that distance they expand and so by the time they hit the detector They're actually sort of centimeters big and so we have a quarter object that is on the centimeter scale by the time it hits the detector That's amazing. And when you say blob what what kind of structure because in a gas It's almost random. Like, you know, they're just sliding around everywhere But you know in a structure sometimes it's like a lattice or but so what does the quantum state look like?

Yeah, how do they look like to compare to each other? Yeah, so because they're all identical They're essentially an identical particle. So it's basically just one one state that just looks a very smooth Smooth sort of blob essentially There's no random motion. So they're not not really bouncing around off each other. There's just essentially one smooth blob I remember first reading about this many years ago and was just totally impressed and it's got Einstein's name on it and Bose was an Indian physicist, right? And we it's we credit him for the word we use for bosons the particles. Is that right?

Yeah, exactly. So so the particles we cool down a bosons We can also cool fermions and they do a whole completely different set of physics But yeah, um, so that there's the two types of elementary particles of bosons and fermions and for Bose-Einstein condensates We use bosons, which yeah, we're named after I'm the Indian physicist Bose who came up with the statistics to describe them and then to describe this state of Bose-Einstein condensation No, Einstein was not a fan of quantum physics. So why did he get something named after him?

What's up with that? Yeah, I mean, it's kind of funny that that Einstein he essentially was one of the inventors of quantum physics It was his his black body radiation paper that That came up with a lot of the initial physics stuff What he had a problem with was some of the aspects of physics and kind of the interpretations of quantum quantum physics We're what he really

Struggled with a little bit some of the things such as entanglement So he didn't like the fact that when you have an entangled system if you have two particles It essentially means that if you take two particles and you separate

them if you measure one of them you'll instantaneously know the state of the other and This is what Einstein didn't like because that implies that something travels faster than the speed of light which famously violates one of his other famous works on relativity and so he really didn't like that and

Yeah, so we worked hard to sort of say that well quantum mechanics must be incomplete There must be a way that there must be some information that we're missing here and it was only sort of Maybe 30 40 50 years after that that we're actually able to prove that no that really is how the world seems to work Yes, so I was wrong in his assumption that it's incomplete. I guess or maybe it's still incomplete Philosophically, but everything works. So no, right? He was just upset because it made him look they made his suppositions

He was like

This can't be the case because now everything that I figured out in my physical representation of the universe can't be so Guess what? No, no The Chuck account of the

So so Sean, but where does entanglement come in to this Bose Einstein condensate? The Boson state condensate we essentially just use as a source for our experiments on entanglement So what we do is we take we take our condensate we split it in two and we collide the two parts of the condensate together So we basically give give one half of it a kick with a laser beam And then it collides with the other half and in those collisions You get all these individual pairs of atoms from each of them will collide off each other and they could go in all sorts of different directions

Let's just focus on two of the directions they could go So if you have two atoms, I can probably do it with I normally do this with a couple of coins So if you have a couple of imagine these these coins at the atoms you bounce off each other They can either go that way or they could go that way. Okay? So again, there's this way or that way and classically if we think of them as little billiard balls They could either go this sort of up or down. It doesn't

they do either of those and You do the experiment they go this way you measure it or they go this way and you measure it now quantum mechanically That's not what happens. What happens quantum mechanically is that when they do the collision They go this way and they go this way at the same time for those who might be listening without the benefit of you too the this way and this way is kind of Northeast and northwest instead of northeast and southwest

And that that's what you're saying quantum mechanically both of those pairs Well, the atoms go both ways However, when you measure it, there's only still two atoms in the system. We're not we're not creating matter here We're not creating anything and so when you measure it, you'll only either get Northwest southeast or northeast southwest But until you measure it the atoms have gone both paths and that's that's what entanglement the entanglement

Oh my god Okay, so now I've only ever heard of entanglement with regard to Particles and now you're describing an entire atom At those temperatures the hydrogen nucleus will have its complement of electrons So you've got a whole freaking atom here and are you saying you're entangling atoms?
Wow Yes

Most entanglement so far has been done with things like photons, uh, which is a very elementary particle. It's just a excitation of light, but like like you said Neil, I mean Helium atom is quite complicated. It's got a nucleus which has two protons two neutrons It's got two electrons whizzing around that and now we're taking two of these and we're entangling them together First of all, that's crazy

Based on everything that you know, so far

It's time to see where Einstein was coming from. Yeah, I am Oh, you're a good company jack. Okay. Yeah Yeah, I mean that's kind of insane You know so now All right, here's how long How long do they stay entangled because that's a big contest out there, right?

I mean, I remembered reading some papers about what they were doing in china Where they had two entangled particles one was in orbit and what was in a lab And so it's like how are they doing this and what are they after? Is it distance or is it time? Or do you have to make sure nobody messes with one of the particles? So you have to you have to make sure it stays isolated. What what are the conditions to sustain this? Yeah, that's that's pretty much all correct. So, um, you can for

So in our case, we're entangling our particles in momentum that or or in the path that they travel So the direction they go, um And that's that's really hard to keep your particles entangled because if they go slightly differently from that path So instead of going northeast southwest, they go slightly towards north northeast say then You're no longer going to be entangled because you've you've now gone on a different path and you'll you've broken this the symmetry of it I guess, yeah, you might you might you might collapse the entanglement or degrade it to an extent

um, where so Our particles don't stay entangled for long. It's it's our whole experiments about a millisecond

That that we do this in entanglement for a millisecond is not a millionth of a second Even though it sounds like it should be Millisecond would be a thousandth of a second, correct? Thousandth of a second exactly. Yes, right like a millimetre is a thousandth of a meter So millisecond is a thousandth of a second. Okay, so I mean that's a very long eternity on a quantum scale

It all depends on your perspective Yeah Is is there a long-term goal for this experiment or is it sort of an existence proof that you can entangle atoms? Yes, so this particular experiment that we did it was um, Uh, Kind of a demonstration that we could do this. So people have there's been a lot of entanglement previously done with photons

There's been some entanglement done with atoms as well Um, but the entanglement that's been done with atoms hasn't involved external degrees of freedom Um, uh by external degrees of freedom. I mean basically the fact that it moves in different paths Um, so things like momentum Previous experiments have just been things such as spin. So you might put it in a superposition of being in different states But they stay at the same place and so ours was the first experiment that showed momentum entanglement with atoms

and the reason why that's interesting is because Uh, one of the things you might want to look at atoms for over photons is that atoms interact much more strongly with the gravitational field And so potentially we could look at effects such as how does, um, gravity interact with entanglement? Down the track we're talking about and so that might open up avenues to explore things such as quantum gravity theories

It's not that gravity doesn't interact with the photons. It would just be much harder to measure Right. So with whereas a tangible particle You've got something whose path you can track. I guess is that is that what's going on there? Yeah, exactly. It's a much stronger interaction

I'm Joel Cherico and I support star talk on patreon. This is star talk with neil de grass tyson

Hey, so can you help me out as I'm trying to wrap my head around this?
Um with the undetermined state um And the path because you said If it if it goes on a different path Can you talk a little bit of more about that because I'm I'm not quite understanding the

You know the superposition Before the actual measurement because if china has a particle in orbit and a particle in the lab That sounds like they're on different paths. So What what where does the sensitivities come from? For changing what one particle does relative to the other? Well, thank you neil. That was my question. Sean

In our case, it's the fact that it's the momentum states of the atom. So it's the direction It's traveling that is the entanglement. So if if if you imagine sort of north south Sorry, northeast southwest northwest southeast

Pairs then um, you can imagine that uh It's it's it's the fact that the atoms are either going northwest southeast or they're going northeast southwest and They're going those two directions at the same time Now what's happened with previous atoms is you'll have an atom sitting somewhere And then you have another atom sitting somewhere and you'll use photons

To communicate between those two and to flip the atom into a particular state. So it's still sitting there. It's doing whatever it's doing It yes, it might be orbiting earth. It might be sitting in a lab. Which is of course Rotating this earth. So it is moving but it's not the motion that's entangled that motion does a change So the in momentum is really the the key here. I got it. Okay I got right and the photon is your measurement Of the state right because you can't measure it unless you interact with it in some way

For our atoms is actually slightly different. So no, I meant for the other case The other case. Yeah, yeah, okay. Got it. Got it. Got it. Okay All right, so this is so so I have my notes here to inquire with you about The bell inequality theorem is that how's that relevant to what's going on here? Yeah, so historically if we're going back to um, I'm shine and what he didn't like about entanglement So he didn't like the idea that you could be in these two states at once and then measure

Um, and that would collapse your superposition so that you'd then know Was that or was that not his invocation of the phrase spooky action at a distance? Yeah, that was exactly what he said So he described it as spooky action at a distance and he says you can't have this you can't have this that You'll collapse to being in this state Um, but before you collapse it you're in both states once he didn't like that. He thought there must be something

And am I correct? In that because I've heard two different answers, but you're the horse's mouth here that the other particle Knows of this not simply faster than light, but instantaneously Which of those is the right way to think about it to the best of our knowledge? It seems to be instantaneous, but it's very hard to prove that it's exactly instantaneous Wow

People have proven that it's faster than faster than our speed of light though So tell me about the the inequality theorem. Yeah, so um Einstein and his co-authors bedosky and rosen wrote this paper Which was the famous spooky action at a distance paper saying that this can't be how the world works and there must be something in quantum mechanics that's incomplete and Everyone for sort of decades kind of thought well, you're never going to be able to test this so

Uh, it's just a philosophical debate to an extent and then fast forward sort of 30 years or so into the 60s and john bell Um, there was another famous theorist uh came up with a experiment where you could actually measure this and because Because the problem is with entanglement if you imagine that you're in this Superposition of northeast southwest and northwest southeast But you only ever get one result out of the system So it's very hard to prove the difference between being in that superposition and not being in that superposition

And so what john bell said is hang on we'll take that superposition and we'll then interfere it back with itself So if you imagine if you have the particles going, um, northeast southwest northwest southeast You then take those two parts of the superposition and you reflect them back on each other So that you have particles that go like this and particles that go like that and you end up There's a spot where they overlap again And because the halves of the superposition can overlap

You can then get quantum interference at that point and so bell came up with this Inequality called the bell inequality, which he said that if they really are in this superposition, you'll get this interference and um, he As far as we know thought that his inequality would always hold Um, it would that *i.e.* the classical physics would um be correct and that um the quantum which predicts that his inequality is violated

um Wouldn't be correct and Then yeah a couple a couple of decades later. Uh, some physicists, uh, such as along our spay and uh, co Um, uh measured it and showed that it is he's actually Violated so so that was a thought experiment not an actual experiment that he conducted. Wow Right, right at that. Let's go to our fan base our patreon supporters

Each paying five dollars a month to gain access to our guests in the form of a question Let's start with john maier And john maier says dear doctors tyson hosman and lord nice I've been reading about entanglement for decades, but I have never felt I could understand its nature So thank you for this episode. My question is three parts So one how do you entangle a particle and two? How do we know they are entangled with spookie etchin?

and not just similarly related So I love the show and bravo And let me jump in the middle and ask can you just take two random particles? And forcibly entangle them or must they be birthed together to be entangled in the way you describe?

um, so you you need to just just to meals. I'll go to nil's point first. So um You just need some way that those particles can be identical So they don't in our case we used identical helium atoms But you wouldn't you wouldn't need to for instance if you had non identical particles you could just collide them off each other and

One would go one way the other would go the other way and they've been a superposition of Say, let's let's again. We'll go back to my coins. Say we have sort of a red and a blue particle You could collide so red goes one way blue goes the other way or blue goes one way red goes the other way You could entangle a red and a blue particle Yeah, yeah, I did not know that I thought they had to be kind of Sort of symmetrically identical, you know with just complementary

Uh Elements like spin or whatever. So I did that's interesting to me. Okay. So in practice, how are you entangling particles? That's the first question right chuck. Yeah, yeah Yeah, yeah, so in practice that's that's um, we we take one One particle and we give it a kick and we collide it with the other particle and they bounce off each other and

Then you get in this superposition of um, the particles going as we're talking about northeast southwest or northwest southeast and um Uh, so you're in these these different momentum states and that's your initial entangled state So the act of colliding them off each other one another entangle Brings their wave functions into harmony So that what was two wave functions becomes one?

Yeah, yeah, I think that's a really good way to describe it. It's it's there's two separate wave functions that you can describe separately Um, and then after you collide them, there's no way that you can describe them with two different wave functions You have to use a single wave function Wow, okay Yeah And this and this is because the wave particle duality of nature at those smaller scales

This is quantum physics at its finest, right? Yeah, okay. So the second question was what chuck How do we know they are actually entangled with spooky action and not just similarly related that seems to be the Uh, the real question there, uh Yeah Yeah, so I think that that kind of goes back to again if they were classical particles you could put them in this

You could put them in kind of a superposition where they either go one way or the other way But quantum mechanically they go one way and the other way at the same time And so the reason we can and the way we can prove that Even though we only have a measure one outcome Is that if we then take the two halves of that superposition and we combine them back together and interfere them Then we can show that there'll be different

outcomes and so the particle can essentially interfere with itself and um We can show we'll get different outcomes to what you would get classically Wow, man That's all right. That is some freaky stuff. I love it. It's freaky friday. Yeah. So this is hayden Gorinj, I think

Gorinj, he says hey dr. Tyson Dr. Hodgman lord nice. Hope you are all doing well It's in the paper by dr. Hodgman at all that the helium atoms were Momentum entangled and I hadn't heard of sub types of entanglement before what other types of entanglement are there And why did the team opt to use momentum entanglement for the b.e.c

Made of helium atoms all the best hayden from london england pip pip What is the inventory of Entanglements that you have Yeah, so there's there's lots of different types of entanglements. So if you have the original experiments with photons It would often be something such as polarization So polarization is just if you imagine a photon as a small particle of light

It's basically which way is the light vibrating is it vibrating this way or is it vibrating this way? So vertical or horizontal and you might entangle your photons in that so that one of your photons has vertical polarization One has horizontal and so you could be in a superposition of vertical going right Horizontal going left or horizontal going left vertical going right and then you could measure that and

Yeah, and so that was a lot of the original experiments were Done with photons with things such as polarization um atoms previous experiments have done things such as spin so spin is just a it's a fundamental property of atoms or

And charge particles that you can have sort of spin up or spin down And you could be in a superposition of spin up and spin down and entangled with that and in our case We did momentum um pretty much any quantum property can be entangled You just have to be able to put it into a superposition of those states Well, let's get back to the spin so In the two entangled spin particles

Does one have to be spin up and the other has to be spin down? No, no, definitely not um as physicists We tend to use a shorthand that if there's if there's any system that is a two spin system um You'll call that spin up and spin down because that's how we tend to learn about it in uh Undergraduate physics and we tend to just as they do. Yes. Yeah. Yeah. Yeah. So for instance

In my lab, we've done previous experiments where we've used a spin one and a spin zero state of helium and we've entangled them um And uh, but we still call it spin up and spin down because it's just it's shorthand that physicists understand easily Because when I think of quantum particles, I think they have complementary quantum states But that's not the case. They can have any quantum state

You just have to be able to couple between those quantum states if you can't couple between the quantum states Then you can't uh interfere them. So if you can't change the coupling just means changing So if I can't change controllably from spin one to spin zero or spin up to spin down or horizontal to vertical polarization Whatever your entanglement parameters are in our experiment. It's momentum um And so if as long as you can change coherently between those states you can

Get entanglement and show that you've uh a show of bell inequality violation Okay, wow Look at that. That is mind-bending stuff. I did not know that I did not know that. Okay Bring on some more. He says hello, dr. Tasman dr. Hosman. This is uh, jonas williams From new york city

My question for you Has the study of quantum entanglement and subatomic particles changed or influenced how you understand reality

I like that. Oh, I like that If you're not purposely entangling particles, is that something that can happen to them by natural causes? Oh, definitely. Yeah. Yeah entanglement can happen. Um all the time It's just that it's Because it happens on such a small scale. Um, we normally don't see the effects of it. Um, and so yeah Yeah, and and I think that so back to the question for me. Um

It's kind of it's really weird and I think a lot of us sort of studied quantum mechanics when you first encounter it whether it's In an undergrad or high school or um through a podcast like this when you first hear about these quantum features You might think yeah, that that that sounds sounds really weird. That can't be how the world works and then Because it's not how it's not how our brains have evolved our brains have kind of evolved to um

Be used to sort of I growth throwing throwing things at animals. I guess, um, which is a very classical way of looking at the world and if I throw a cricket ball or what do you have baseball in america if I throw a baseball, let's say, um, and It's not going to be in two places at once. It's not going to go two directions at the same time And so when we read about quantum mechanically that a small particle, which we like to think of as just a smaller smaller ball um

When we read that can go both directions at once Uh, we just sort of think oh well, it must be some mathematical trick and The fact that we can actually do these experiments to show no no at the very small scale or very cold um This is how the universe works. It's yeah, it's a bit mind-bending, right? Yeah without a doubt to summarize

Because it happens on these microscopic scales. There's no macroscopic manifestation Of entanglement that we experience Yeah, that's because we're just trying to not get eaten by the lion and that doesn't require quantum physics to accomplish Are they forever forever? separated the quantum and the macro physical or is there a possibility that there is a

point of crossover Yeah, that's that that's a really good question and that's that's a really active area of research at the moment that Because clearly at the small scale quantum works at the large scale quantum doesn't But there's got to be somewhere in between. There's either must be a hard point where they stop working Or maybe it's a fuzzy boundary. A lot of the research we're kind of seeing these days Is that there might be a fuzzy boundary where you just kind of get less and less quantum and more and more classical

Wow. All right. God, how do you deal with this every day, man?

All right, here we go. This is do you arrive at home at the end of the day depressed or jubilant Normally depends on how much university bureaucracy I've had to deal with in a particular day Quantum physics is the easy part of the job sometimes

As they say the administration is a new kind of subatomic particle called the moron Oh Damn, okay, this is mike parker and mike parker says dr.

Hodgman dr. Tyson lord nice Mike parker from virginia. It seems entanglement is observed in larger and larger particles Is there an upper-sized limit to entanglement? Could there ever be a way to use particle entanglement for instant long distance communication or other

Practical purposes. This is the question that every sci-fi fan wants to know man Can we have faster than like communication like on star trek? where Captain there's a subspace communication waiting for you in your quarters

Or can you entangle molecules? I mean, yeah, we're impressed with the atoms, but whatever you've done with us lately Can you can you entangle molecules? Be they simple or complex on the Entangling larger particles. That's that's another really active area of research people keep pushing it further and further There have been people that have entangled collections of atoms and molecules and they sort of push out I believe I believe it's of order a thousand atoms, but don't quote me on that because I could be um

I could be wrong on that. Well, it's all I care about. Yeah, it's way more than way more than one. Yeah Yep, and so yeah And trying to push this entanglement on the bigger scale and work out where it breaks down at some level. It clearly doesn't Uh, it doesn't work anymore. And so yeah, where that transition is is a really active area of research So maybe you were familiar with george gamma's mr. Tomkins in wonderland I don't know if you I'm a little older than you so I don't know if this was around in your day

but george gamma Mr. Wonderland was a place where the laws of physics were different simply by The physical constants being different. So one of them the speed of light is like 60 miles per hour Right and so you're driving down the street and then you see these these relativistic effects just by Approaching the speed of light as your speed limit on the road one of them was they changed the value of the Planck's constant so that

macroscopic objects would feel Quantum phenomena that you'd walk through the door and you'd like diffract

It's fun to think about what how that if you have a knob that could tune it What different things would happen to you or we just simply develop a new

sense of Of what the world is Right, like you said, we throw the ball and it follows one arc but if quantum rules We're all around us at all times

And you threw the ball and it split into two states You would tell it's just splitting into two states. It wouldn't even be odd To watch that because it would happen so frequently. Is that a fair way to think about this? Yeah, I think so our brains are great at adapting and i'm pretty sure that if if we'd had to evolve to live in a quantum world I'm sure we just think it was normal Yeah, it's just normal. That's right. That's right. And what about the communication portion of mic's question?

Oh, yeah, yeah, yeah, yeah, yeah, would it ever be a viable use for long-distance communication? Is there a future in this? Yeah, so this this is what everyone thinks as soon as you hear entanglement you think there's instant Measuring one instantly changes the other we could use that to communicate faster than the speed of light um, unfortunately, that doesn't seem to be the case And essentially it's because the way you measure it

Um contains information. So if you make a measurement on that particle Yes, you'll you will know what the other one is but you won't be able to communicate that in a useful way unless you communicate it classically because You'll let you'll then the other person would then have to Make a particular measurement to get anything useful out of that and so um, yeah, you wouldn't you can't actually use it to For faster than light communication. There's there's a theorem called the no communication theorem Which basically says that yeah, you can't um you can't use entangled states to communicate faster than the speed of light

Really? That's the name of the theorem the no communication theorem. That's the best name they can come up with

I believe that's what many women call their husbands

The no communication theorem But yeah, so so basically what you're saying is the collapse of the superposition and Once that information is set You you can't know it unless you're making the same measurement that the other person is making on the other side And you wouldn't be able to know which position it is unless you were to call them and say here's the position

Yeah, it's exactly because because quantum measurement perturbs the state um if you If you make your measurement in a way that uh could exploit that information It will change the state of it And so then that will change the state of it and then when they make their measurement They'll get a different result out of it

Unless you've called ahead and told them how you need to make this particular measurement Which by the way is how you know that it's entangled That's how you know it's entangled Exactly. It's kind of frustrating. You're like we should be able to do this, but yeah, we we can't so we can't use it for

Encrypted message sending Uh now that's that's a completely different question. You can use entanglement for encrypted message sending So that's that's where you exploit the fact that if you measure one Half of that you will change the other half and so if you encrypt your message on two uh Photon pairs and you send one of them to the person that you want to and you keep the other one yourself

Then if you make measurements on that and compare the results With the person you've sent it to you'll know if anyone's messed with your system because you'll get different results and so you'll know if it has been eavesdropping on your messaging and so there's there's mathematically provably secure encryption protocols using quantum communication which can be shown that

You just can't break them because if you interfered with it If you you did a fear with the state if you listen to it you interfere with the state and people would know your eavesdropping and um You could just abort the communication So this is a pipe dream then it's a pipe dream that people have for it these things have been demonstrated um at at at various scales and um, yeah, they're they're essentially um, It's it's almost an engineering problem at this stage

to get it to work better Okay, look at that. All right. This is william warren and william says Hi, dr. Tyson and dr. Hodgman. This is william warren from abingdon, maryland If quantum entanglement is a fundamental feature

of nature could spacetime itself Emerge from a vast network of entangled particles in other words Is it possible that distance isn't fundamental?

But rather a consequence of how information is connected at the quantum level Thank you so much we've heard on another installment of star talk from one of our physicist friends brian green That it might be that the virtual particles in the vacuum of space

that are connected to each other by entanglement they're entangled that that entangled Gap between them may be

A wormhole and a wormhole would have the same property because You just step through and it's you're just there, right? You're not Moving faster than light the the hole enabled that and it's not because you had special rockets. So so what What is the latest thinking other than what I just shared with you about what the entangled pathway actually is? Yeah, wow, this is um, you got to remember. I'm only a dumb experimentalist. Um, I basically

Experimentalist, okay I probably I would certainly defer um Yeah, thinking on that to things such as to the experts like brian and um and code because um Yeah, that's that's kind of a bit a bit beyond. Um, what we're working on. I it certainly sounds like a um Uh, interesting take, uh, but I'm yeah, I'm not not really sure. Unfortunately. Sorry

Okay, no, okay. But listen we we like that answer

You know now is this hard to get somebody to say hey, I'm not that sure. Sorry, you know what I mean Yeah It's really good point I think it's a big it's a big part of being a scientist There's you've got to learn to know what don't you know and it's one of those things that the more, you know The less, you know, you know, it's kind of the reverse dunning kruger effect

You know the dunning kruger effect that the less, you know the more you think you know It's kind of the reverse the more, you know the more you realize that wow, there's heaps of stuff that I just Don't know at all and um, yeah Dunning kruger we get along very well very well Dunning and kruger. I know them. We have a great relationship. They told me I do the best

dunning and the best kruger All right, I think there's a I think there's a value in

Saying I don't know and allowing people to understand that that science is sometimes the answer is well, we don't know You know The two different the science doesn't know and there's this I don't know those two different things Right, and they're both the same from me Oh

Whether physics itself knows I think the answer's still out on that as well I think there's still um some open theories in that because again You can come up with a theory But you have to be able to prove it experimentally and I think to prove something like that experimentally It would be extremely hard So there's there's some really interesting theories and the trick is going to be a bit like with einstein spooky action in a distance claim

Coming out with an experiment to test it. That's that's what we'll be. Yeah, that that would be a big Nobel prize discovery cool All right, let's go to alejandro guardado And he says

He's from hack and sack new jersey Where's he from? Where's he from? Uh, I don't think alejandro says where he's from Oh, okay He says No, and washington state. I'm sorry. He did say washington state. Washington state speaks that way. I just want you to know Okay, all right

That's that's my fantasy of alejandro in washington's in washington state um He says hey dr. Tyson dr. Hodgman lord nice alejandro here from Washington state my question is what stabilizes quantum fluctuations when particles fuse or collide how does this Increasing proximity effect entanglement

more extremely How would this work for particles in the singularity of a black hole? Thank you Yeah, so I want to reshape that just a little bit. So if I have two particles, we know the quantum physics says there's Everything is always in motion. There's always some energy to the state

It does that disrupt any attempt to Entangle two particles. So in other words can two particles natively break apart Simply because of quantum

fluctuations that are inherent in all particles in all systems quantum fluctuations um At some level will probably have an effect on entanglement

But it's normally at such a small scale that it won't stop entanglement happening I think and in fact when you cool down your helium atoms You are reducing the quantum fluctuations by dropping the temperature, correct Uh, we're kind of reducing the classical fluctuations. So temperature is temperature is really a classical a classical Phenomenon, um, it's just random motion of particles. We're reducing the classical fluctuations to get to the scale Where you can in principle see quantum fluctuations. However for our entanglement

Um, the quantum fluctuate for our particular system. The quantum fluctuations don't really come into it It's there's there's much larger things that cause the entanglement to deco here things such as our classical so magnetic stray magnetic fields and um Yeah, the like your microscopic disruptions to it. Yeah, okay. Yeah, definitely We're not really at a level where we're sensitive to the microscopic quantum fluctuations This is

Mikael boysvert who says Hello guardians of the geeks Mikael here from canada. I like that guardians of the geeks, right? How would the universe change if the programmer behind would suddenly toggle particle entanglement off?

We know life By the way In the same spirit of that I heard someone suggest that evidence we're in in a in a simulation is that the programmer already put a limit To how fast things can go because they can't

Simulate it faster than that. So the speed of light is the programmers limit That we've bumped up against And it took a long time to get there, but we finally got there. So it's like in the Truman show He finally gets to the the outer Uh edge of of the set of the telemetry of the set of the set So

Yeah, so what happened to our quantum entanglement? What's different about the world? Yeah, so at the macroscopic level, I think um Probably probably not not a huge amount like that entanglements only at a very small scale. Um, however Well, yeah, however, there are a range of processes that we're starting to Have hints at that may be really entanglement may be

really important. There's some biological processes such as um People are postulating the stability of DNA may be due to um entanglement. It was in the molecules itself. Um

Navigation of birds some birds use magnetic sensors and there's hints that um, there's quantum elements of that that rely on entanglement Migration migration of birds migration of birds. Yeah navigation during migration is kind of what I mean Like like when when when they migrate, how do they know to go north? Yeah, how do they know which way to go? And so there's there's there's there's thoughts of that and even things such as I think photosynthesis is the latest one that they're looking at that may have um elements now

I think the jury's still out on all of these again, it's not quite my area of expertise but I believe the jury is still out on all of these as to whether it's um Definitely quantum entanglement enhanced But there's definitely some evidence that's starting to point towards some of these biological processes Quantum physics is really important and entanglement in particular. And so Yeah, well at first glance you might think that if the programmer of the universe turned off entanglement

We wouldn't see anything any distance. Uh, it may actually be uh, really important Okay, very cool. I love the idea of photosynthesis and quantum entanglement. That sounds so cool All right, this is bruce Lessie and bruce says dr. Tyson's and hodgeman. This is bruce lessie from cripple creek, virginia my question is related to entanglement and spooky action at a distance as Einstein phrased it if space time for a photo traveling at the speed, uh, he said photo, but I think he means photon

Uh, traveling at the speed of light represents zero time experienced by the photon Because at the speed of the light Distance is non-existent. Why did Einstein have a problem with instantaneous communications between entangled particles? Nothing is at a distance for particles that travel at the speed of light Yeah, so I'd I'd say that's one of the reasons why it's really important to measure entanglement for things not just That aren't photons. So yeah, like like because photons travel at the speed of light

If you just measure entanglement with that you could perhaps come up with some sort of explanation like that But in our system, we measure it with atoms our atoms move very slowly We give them a kick and they move at several centimeters per second. Let's call it an inch per second for your americans with your freedom units

Freedom units, that's exactly what we use to measure the octagon on the white house God

So yes, so back back to the atoms so it because they're moving Relatively slowly you can't come up with that sort of an explanation to explain it. Um, it really uh And simultaneous spot this and yeah instantaneous communication suddenly becomes a problem again well, that's really cool, man because yeah, they have mass and um Yeah, they're entangled at much much much slower speeds

So, yeah, that's really that's what well, it's still a good question That's a great question. Great question. Yeah, what really good question bruce. Thanks. All right. This is david barlow and david barlow says Greetings, dr. Hosman dr. Tyson lord nice david barlow from chicago illinois here a newly signed up patreon supporter kudos to you my friend He says I was wondering if As an experimental physicist and an observer within the quantum field

You and your associates knew for a certain for certain that you were not Affecting the results of your experiments in the bell's theorem What loophole prevention precautions were taken to negate your field collapse of the wave function? When taking measurements love your fantastic science broadcast guy So, so yeah, how you know this that you're not messing it up like

It could work me that's that you always have a big doubt that yeah, did we just do something wrong when measure something That's why we have to do sort of rigorous and multiple tests and um And and for the scientific method in general It's part of the reason why you have to publish these papers that other scientists can then go hang on Did you think about this? Did you try turning this off? Did you try turning that on and yeah, um, it's

it's yeah It's an important part of the process. Um for our particular experiment

So maybe I should just briefly cover what loopholes are that are referred to there So with with bell with bell experiments. Um, there's these things called loopholes, which are basically Ways that people come up to say. Well, maybe to still preserve localities to still preserve the fact that um You don't have instantaneous communication and so it can be things such as well. Maybe Uh, maybe the experiments conspiring in a way such that it only lets certain results through or maybe your um, uh, maybe the observer is is uh, doing something is communicating away

Maybe you look like like like if you if you set the interference on the true halves of the system and the measurement um, if you don't set them after you've If you set them in a way that they could communicate with each other Maybe there would be some really weird um theory that could explain that um, and so uh, a lot of these all these loopholes have been closed with the um

Experiments on photons for our particular experiment. Um, we didn't close all these loopholes. So our experiment isn't loophole free in principle Um, you could make some of these criticisms about our experiment However, the fact that they've been closed with photons means that we'd probably expect that they should also be closed for atoms as well um, and so yeah um And part of the reason why we didn't close them was because a couple of them are technically extremely challenging to close with atoms

Just because atoms move a lot slower than photons and they move over much smaller distances And so yeah, it's an ongoing area of work Where it would be a good area of future work for groups like us That's a damn good question. Chuck, we have time for a couple more Cleo fox. He says hello, dr. Tyson. My name is Cleo from denver colorado Modern quantum physics has achieved extraordinary predictive accuracy

But many of its foundational interpretations remain experimentally indistinguishable Given recent advances in quantum information theory weak measurements quantum computing and tests of non locality Do you think the next major breakthrough in quantum physics is more likely to

come from developing new mathematical frameworks? Or from entirely new experimental Methodologies capable of probing quantum phenomena in ways we cannot currently access put another way

Are are are we currently limited by our theories or by the tools we use to test them? Yeah, so I was gonna say something similar to that in summary of that question. So Sean there's, you know philosophers like believing they have access to

Emerging truths In science in general, but especially in quantum physics where there's so much that makes no freaking sense So is there room for philosophers to guide the physicists? Through this and or out of it Or we just stuck just as they say shut up and calculate And so that's that's a that's a nuanced way of saying

Is it what was the final question there? Are we currently limited by our theories and the tools we use to test them? On on the shut up and calculate versus philosophy debate I'm a big believe as an experimenter if I'm a big fan of shut up and calculate experiments are hard enough ideas Count me on that vote as well. Okay Yeah, it's basically the maths tells us the results that give us predictions of our experiments

And it works. Yeah, and it works and and and so at that level Yeah, I'm happy with that. I'm happy to leave the questions to the essentially to the realm of philosophy to an extent if if you if you're um If you can't actually make predictions of what an experiment will give you I think that's basically philosophy and um I think there's definitely a role for that. I mean we've we've seen that how brain bending some of these quantum effects are

And I think it's really interesting to probe that I think it's really interesting to have philosophers and the like guide that and theorists and interact with quantum theorists But I think it's also equally really important for experimentalists to actually test these results And if you can if you can't test these results, then You probably need to work harder on your theory. I think

I spent some time at Princeton where they they're very theory based although they do quite a bit of experiments they are in the They have a tokamak and a fusion reactor, but there's a strong theoretical legacy in the

department And I there's a sign up somewhere or someone someone's door
It says never trust an observation unless it's backed up by a good theory

Yeah Yeah, so I think I think coming back to the question it really is it's
both There's there's a lot of work on theory, but there's also a lot of work on
experiments to cover that Do they do they inform one another? I think is
kind of also the the spirit of the question Deally, there you go. That's That's
the correct way to think about that Chuck

Absolutely, there's there's plenty of times like our experimental
entanglement that was originally proposed by some theorists We tried to do
it the way they proposed and it didn't work So then we came up with a
slightly different way and then we came back to them and then they helped
analyze their results And yeah, it's really there's a lot of back and forth
Okay, um, cool Cool. I chuckled one more question. We got time for

Uh, right. Let's close it out with melanie stickler And melanie says hello, dr.
Tyson. I'm melanie originally from austria now in the bay area of california
My question is about dr. Hodgman's helium experiment. I understand
massless photons have wave particle duality But how does a massive
particle like a helium atom which is subject to gravity? Function as a wave
cloud Furthermore, if measuring a quantum system collapses the wave
function, how did the team measure the atoms?

in simultaneous momentum states without Instantly destroying the
superposition I'm sure the paper covers this, but I'm having a hard time
wrapping my head around it So thank you. Yeah, helium is a massive
particle Compared to stuff we're used to And so there's a wave function
associated with such a massive particle. Yeah, definitely and um,

It's a great question. And um Yeah, the there's a wave function associated
with helium atoms And that's why we need to cool them down to make it
work Because otherwise at room temperature the wave function is so small
that you can't see it But at these temperatures the wave function is
macroscopic It's sort of in the order of tens to hundreds of micrometers. So
that's a point one of a millimeter and um

Yeah, it's quite large at that scale. Um, the other part of the question if I remember if I'm getting it correctly was How do you measure that they're in two different states at once when they're only ever going to be in? How do you prove they're in two different states at once when you can only have a measure one result? And again, that really comes down to John Bell's work for how you can measure this Bell inequality Where if you interfere those states you can measure the results of that in the outcomes you get you can

Interfere those states and you can get more probability of being in one than the other In your output due to the fact that you were in this superposition of two states at the same time Is there a quantum entanglement arms race in the world? Like who's leading the quantum entanglement? Experiments because I don't think it's us. Is it us Americans? You're in Australia. Who's who's ahead of who's ahead and who's behind? Um, it kind of depends what you're talking about. I mean our our experiments probing the fundamentals of entanglement Um, uh, there's there's there's still work going on in that

But a lot of what current quantum research is going into is how we do something useful with entanglement So how you can use it so things such as quantum computing. So quantum computing Is a computer that rather than using bits to encode information So bits have to be one or zero you use qubits which can be in a superposition of one and zero at the same time And then if you have multiples of these together you can end up with you can entangle the different qubits And by the fact that you can have qubits in many states at once for certain types of problems

You can probe Many answers at once even though you only ever get one result when you get out and so when you do your final measurement and so There's a lot so in principle. It seems really promising that you should be able to do Uh, much faster calculations much higher level computation with this But the problem is that finding

It only there's only particular problems that we know of that. This is true That's so there's there's a lot of work going into that Then the other problem is that quantum systems are really hard to get to work on a large scale So I have lots of qubits and so there's a lot of work going into

building these processes And so yeah, there really is if you want to call it an arms race. There's a lot of Government and and private

Investment in this at the moment. There's a lot of quantum computing startups all around the world on the assumption that whoever gets advances in it first might have a leg up either economically or with regard to security or Computing and so Even if at the end of the day It's just a pipe dream because it's like it's just a fun physics exercise

But it doesn't have any practical use no one knows that yet. Is that a correct way to think about that? Absolutely, it's still it's still really an open question as to what the impact of quantum computing will be quantum computing could be anything from Like you say have massive economic implications massive security implications It could be used to things such as our medical implications like drug development Uh data processing all all these massive things

Or it could just be on a much smaller level and that it's kind of a toy physics system that helps us advance physics But may not have quite such a wide Economic implication and I think it's a really really exciting time I bet people said the same thing about quantum physics a century ago This is just a curiosity on the fringes of physics. We'll never have any use for this It's still fun anyway

And now it's the foundation of our it revolution everything. Yeah Everything. Yeah, it's one of the reasons it's really important to invest in basic research I mean the the work the work my group does is We don't know what that is anymore United States, we don't know what that is anymore. Exactly. We don't we don't do research here anymore. Okay. Will you hire hire hire?

Australia is not much better. Unfortunately, we run on vibes vibes No, right now it's on physics fumes. That's all this life. Oh, yeah, unfortunately. Yeah Yeah Well, sure. That's all the time we have I'm saddened by this because there's this topic has no end of curious people out there

Thinking about it. They've read about it and it's not every day you get to bump into someone who gets paid for thinking about

Congratulations on your Bose-Einstein condensate As a source of entanglement
Okay, we will look for our invitation to Stockholm in the mail really good
and thanks. Thanks a lot for the questions And yeah, thanks for having me
on your show. I am Neil deGrasse Tyson. You're a personal astrophysicist
finishing up a very special edition of Cosmic Queries

specializing In quantum entanglement until next time keep looking up